EEL 5764 | Computer Architecture

Class Number: 12082

Fall 2026

Delivery Method: Online (80-99%)
TH | Period 4-5 (10.40 AM-12.35 PM) Engineering 0100
T | Period 4 (10.40 AM-11.30 AM) Engineering 0100

Instructor Information

Name: SANDIP RAY

Email: sandip@ece.ufl.edu
Telephone: 3523921605

Office Hours
Thu 2-3pm
Preferred Method of Communication: Email

Office Location

Malachowski Hall 4131

Course Details

Catalog Description: Fundamentals in design and quantitative analysis of modern computer architecture and systems, including instruction set architecture, basic and advanced pipelining, superscalar and VLIW instruction-level parallelism, memory hierarchy, storage, and interconnects.
Pre- and Co-Requisites: Prereq: Digital Design.
Credit Hours: 3
Course Fees: $9.00

Required Materials

COMPUTER ARCHITECTURE: A QUANTITATIVE APPROACH
ISBN: 9780123838728
Authors: HENNESSY/PATTERSON
Publisher: MORGAN KAUFMAN

Edition: 5TH

Computer Architecture: A Quantitative Approach, 6E (Kortext eBook)
Publisher: Elsevier
All Access: This material is available on All Access

Recommended Materials

COMPUTER ORGANIZATION AND DESIGN MIPS EDITION: THE HARDWARE/SOFTWARE INTERFACE
ISBN: 9780124077263
Authors: HENNESSY/PATTERSON
Publisher: ELSEVIER
Edition: 5TH

Course Goals and Objectives

This course introduces students to the fundamental principles, mechanisms, and tradeoffs underlying modern computer architecture. The course aims to familiarize students with the major components of contemporary computing systems and to explore how advances in semiconductor technology, architectural innovations, and software systems collectively shape computing performance, efficiency, and functionality.

The course aims to develop an understanding of key architectural techniques used in modern processors and systems, including memory hierarchy design, instruction set architecture, pipelining, out-of-order execution, speculation, and parallel processing. It will explore the different forms of parallelism available in computing systems and the architectural mechanisms used to exploit them.

The course also aims to familiarize students with quantitative approaches for evaluating architectural design alternatives and understanding tradeoffs among performance, energy efficiency, complexity, and other system objectives. Finally, the course explores the interactions between hardware architecture and software systems, highlighting how architectural decisions influence software behavior and how software requirements drive architectural innovation.

Expectations and Student Learning Outcomes

Upon successful completion of this course, students will be able to:

1. Explain how semiconductor technology trends and implementation constraints influence computer architecture design decisions.
2. Analyze the design and operation of key architectural mechanisms in modern computing systems, including instruction set architectures,

memory hierarchies, pipelining, out-of-order execution, and speculative execution.
3. Evaluate different forms of parallelism, including instruction-level, thread-level, and data-level parallelism, and assess architectural techniques for exploiting them.
4. Apply quantitative methods and performance models to evaluate architectural design alternatives and analyze tradeoffs involving performance, energy efficiency, cost, and complexity.
5. Analyze the interactions between computer architecture and software systems, and evaluate how architectural features influence software performance, programmability, and system behavior.
6. Critically assess architectural design choices in contemporary and emerging computing systems using principles learned in the course.

Methods of Evaluation

Your grade will be based on 4-5 homework sets, one mid-term, and a term paper. The weightage of the different components in your final grade will be based on the following table. Homework with the lowest grade will be dropped.

Assignment	**Percentage of Grade**
Homeworks (4-5)	40
Mid-term	25
Term Paper	30
Class Participation	5
Total	**100**

Grading Scale

The assigned letter grade will be based on the following table. The class is not curved. If everyone does well everyone can get A.

Letter grade and percentage

Letter	**Percentage Value**
A	94 - 100%
A -	90 - 93%
B +	87 - 89%
B	83 - 86%
B -	80 - 82%
C +	77 - 79%
C	73 - 76%
C -	70 - 72%
D +	67 - 69%

Letter	Percentage Value
D	63 - 66%
D -	60 - 62%
E	59% and below

Course Schedule

Folloiwng is a tentative course schedule. The schedule is provided roughly for guidance. We will adjust the schedule based on the background and interest of the class.

Module	Duration (weeks)
Technology Trends + Quantitative Evaluation	2
RISC-V and ISA Concepts	1
Pipelining	1
ILP / OoO / Speculation	2
Midterm	1
Memory Hierarchy	2
Parallelism (TLP + DLP)	2
Modern Architectures (GPU / AI Accelerators)	1
Research Topics / Presentations	1

Alignment of SLOs

The course is designed to support the achievement of the stated student learning outcomes through a combination of lectures, homework assignments, examinations, and a semester-long project. Lectures introduce the fundamental concepts of modern computer architecture, including technology trends, processor design, memory systems, parallelism, and hardware-software interactions. Homework assignments provide opportunities for students to apply quantitative performance analysis techniques, evaluate architectural tradeoffs, and analyze the behavior of architectural mechanisms such as pipelines, caches, and parallel processing systems. The midterm examination assesses students' understanding of core architectural principles and their ability to reason about architectural design decisions using quantitative methods. The semester project further reinforces the learning outcomes by requiring students to investigate a contemporary architecture topic, critically evaluate existing approaches, and synthesize concepts from the course to analyze or develop architectural

solutions. Together, these course components provide multiple opportunities for students to develop and demonstrate mastery of the learning outcomes.

University Policies and Resources

Information about grading policies, support for students with disabilities, course evaluations, the Honor Code, and other course policies and campus resources can be found on the Syllabus Policies page.

Attendance Policy

Excused and Unexcused Absences

Students may only participate in classes if they are registered officially or approved to audit with evidence of having paid audit fees. The Office of the University Registrar provides official class rolls to instructors.

Students are responsible for satisfying all academic objectives as defined by the instructor. Absences count from the first-class meeting.

Acceptable reasons for absence from or failure to engage in class include illness; Title IX-related situations; serious accidents or emergencies affecting the student, their roommates, or their family; special curricular requirements (e.g., judging trips, field trips, professional conferences); military obligation; severe weather conditions that prevent class participation; religious holidays; participation in official university activities (e.g., music performances, athletic competition, debate); and court-imposed legal obligations (e.g., jury duty or subpoena). Other reasons (e.g., a job interview or club activity) may be deemed acceptable if approved by the instructor.

For all planned absences, a student in a situation that allows an excused absence from a class, or any required class activity must inform the instructor as early as possible prior to the class. For all unplanned absences because of accidents or emergency situations, students should contact their instructor as soon as conditions permit.

Students shall be permitted a reasonable amount of time to make up the material or activities covered during absence from class or inability to engage in class activities because of the reasons outlined above.

If a student does not participate in at least one of the first two class meetings of a course or laboratory in which they are registered, and they have not contacted

the department to indicate their intent, the student can be dropped from the course. Students must not assume that they will be dropped, however. The department will notify students if they have been dropped from a course or laboratory.

The university recognizes the right of the instructor to make attendance mandatory and require documentation for absences (except for religious holidays), missed work, or inability to fully engage in class. After due warning, an instructor can prohibit further attendance and subsequently assign a failing grade for excessive absences.

Religious Holidays Guidelines

At the University of Florida, students and faculty work together to allow students the opportunity to observe the holy days of their faith. A student should inform the faculty member of the religious observances of their faith that will conflict with class attendance, with tests or examinations, or with other class activities prior to the class or occurrence of that test or activity. The faculty member is then obligated to accommodate that particular student's religious observances. Because students represent a myriad of cultures and many faiths, the University of Florida is not able to assure that scheduled academic activities do not conflict with the holy days of all religious groups. Accordingly, individual students should make their need for an excused absence known in advance of the scheduled activities.

The Florida Board of Education and state law govern university policy regarding observance of religious holidays.

Guidelines

- Students, upon prior notification to their instructors, shall be excused from class or other scheduled academic activity to observe a religious holy day of their faith.
- Students shall be permitted a reasonable amount of time to make up the material or activities covered in their absence.
- Students shall not be penalized due to absence from class or other scheduled academic activity because of religious observances.

If a faculty member is informed of or is aware that a significant number of students are likely to be absent from class because of a religious observance, the faculty member should not schedule a major exam or other academic event at that time.

A student who is to be excused from class for a religious observance is not required to provide a second party certification of the reason for the absence. Furthermore, a student who believes that they have been unreasonably denied an education benefit due to religious beliefs or practices may seek redress through the student grievance procedure.

Absence due to Illness

A student who is absent from class or any required class-related activity because of illness should contact their instructor, if feasible, as early as possible prior to the missed class or activity.

Students shall be permitted a reasonable amount of time to make up the material or activities covered during an excused absence.

Students should contact their college by the deadline to drop a course for medical reasons. Students can petition the Dean of Students Office to drop a course for medical reasons. The university's policy regarding medical excuse from classes is maintained by the Student Health Care Center.

Twelve-Day Rule

Students who participate in university-sponsored athletic or scholarly activities are permitted to be absent 12 scholastic days per semester without penalty. A scholastic day is any day on which regular class work is scheduled as defined in the approved university calendar. More Info

The student or student's advisor must notify the instructor as early as possible prior to the anticipated absence to allow ample time for accommodations. Instructors must be flexible and not penalize students when re-scheduling during-term and final exams, class assignments, and other required activities and must follow the UF Attendance Policy herein and UF Examination Policies. As noted in the UF Examination Policies, during-term exams should be re-scheduled no later than before the end of the semester, while final exams no later than 90 days after the originally scheduled exam time. However, instructors are encouraged to re-schedule final and during-term exams, assignments, and other activities as soon as possible after the last day of the absence and must not penalize the student in any way. More Info

A group's schedule that requires absence of more than 12 scholastic days should be adjusted so that no student is absent from campus more than 12 scholastic

days. Students who previously have been warned in writing by their instructor about the impact of absences on their individual class performance should not incur additional absences, even if they have not been absent 12 scholastic days. The student is responsible to maintain satisfactory academic performance and attendance.

Course Policies and Resources

Students are expected to attend lectures and ask questions. Your class participation grade will be based on your degree of interaction in class.

Please be in class on time.

Interaction via canvas discussion will count towards class participation grade. Edge students can interact in Canvas to incur credit towards class participation grade.

Late and Make Up Work Policy

Late homework submissions will incur a penalty of 10% per day. Late projects or term papers will not be accepted.

Makeup midterms will not be given except for documented medical emergency. If you have a concern regarding the date of the midterm please talk to the instructor in advance.

Technology in the Classroom

Lectures will be recorded.

Midterm is tentatively planned to be conducted via honorlock, subject to approval by the department.